

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary - Summer 2024

Course: BTech Branch : Mechanical Engineering/Mechanical Engineering(Sandwich)

Semester :IV

Subject Code & Name: BTMC401 Manufacturing Processes-I

Max Marks: 60

Date:12/06/2024

Duration: 3hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define pattern. Explain any four types of patterns.	CO1	6
B) What is the function of core? What is core venting?	CO1	6
C) Explain the investment casting process with a sketch. Why is the investment casting process capable of producing fine surface details on castings?	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain any three examples of rolling stand arrangements.	CO2	6
B) Briefly explain about open die forging and closed die forging.	CO2	6
C) What are the different types of defects may occur in extrusion and drawing operations?	CO2	6
Q. 3 Solve Any Two of the following.		12
A) How cutting takes place when an oxyfuel gas torch is used? Also give its applications.	CO3	6
B) Explain the Gas Metal-arc Welding (MIG welding) with a neat sketch.	CO3	6
C) Describe the functions and characteristics of electrodes. What functions do coatings have?	CO4	6
Q.4 Solve Any Two of the following.		12
A) What are the different machining operations that can be performed on a lathe machine? Describe them with neat sketches.	CO3	6
B) Distinguish between the drilling and boring operations. Explain how can we perform both operations on a lathe?	CO4	6
C) What are the different types of chips generated in turning operations? Which type of chips are desirable?	CO4	6

Q. 5	Solve Any Two of the following.		12
A)	What is the difference between end milling and face milling operations? Explain them with neat sketches.	CO5	6
B)	What do you mean by broaching operation? How it is done?	CO5	6
C)	What are the different processes used for gear finishing? Explain them.	CO6	6

***** End *****